Assignment 2 data-110

Import Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Load ALL 3 FILE TYPES

Loading TXT file

```
df_txt = pd.read_csv("/content/Facebook Friends.txt", encoding='utf-16', sep='\t')

print("TXT File Loaded Successfully")
print("Shape:", df_txt.shape)
print(df_txt.head())
```

```
TXT File Loaded Successfully
Shape: (715, 31)
   Unnamed: 0  Unnamed: 1 Unnamed: 2  Unnamed: 3  Unnamed: 4  Unnamed: 5  \
0           1          57          7          27           5           1
1           2          42        531         241          19           0
2           3          42       1396         423          40           0
3           4          55        394         139          48           0
4           5          30        916         231          78           1

   Unnamed: 6  Unnamed: 7  Unnamed: 8  Friend   ...  Replies  MidWest  Seast  \
0           1           1           0        1  ...       68      NaN    NaN
1           0           0           0        1  ...      234      NaN    NaN
2           1           1           0        1  ...      727      NaN    NaN
3           1           1           1        1  ...      437      NaN    NaN
4           1           1           1        1  ...      372      NaN    NaN

   West  Children  Likes  Edu  Events  USA  Friends
0   NaN       NaN    NaN  NaN     NaN  NaN      NaN
1   NaN       NaN    NaN  NaN     NaN  NaN      NaN
2   NaN       NaN    NaN  NaN     NaN  NaN      NaN
3   NaN       NaN    NaN  NaN     NaN  NaN      NaN
4   NaN       NaN    NaN  NaN     NaN  NaN      NaN

[5 rows x 31 columns]
```

Load CSV File

```
df_csv = pd.read_csv("/content/Facebook Friends.csv")
print("\nCSV File Loaded Successfully")
print("Shape:", df_csv.shape)
print(df_csv.head())
```

```
CSV File Loaded Successfully
Shape: (715, 3)
  Friend \tAge\tPhotos\t# of Tags\tAlbums\tGender \tEmp\tProfile\tCover\tOrientation\tRelationship\tPosts\tReplies\tMidWest\t
0  1\t57\t7\t27\t5\t1\t1\t1\t0\t1\t1\t1\t0\t0\t0\...
1  2\t42\t531\t241\t19\t0\t0\t0\t0\t1\t1\t5\t1\t0...
2  3\t42\t1396\t423\t40\t0\t1\t1\t0\t1\t1\t186\t3...
3  4\t55\t394\t139\t48\t0\t1\t1\t1\t1\t1\t6\t4\t1...
4  5\t30\t916\t231\t78\t1\t1\t1\t1\t1\t1\t412\t75...

  Unnamed: 1 Unnamed: 2
0        NaN        NaN
1        NaN        NaN
2        NaN        NaN
3        NaN        NaN
4        NaN        NaN
```

Load Excel File

```
df_excel = pd.read_excel("/content/Facebook Friends.xlsx")
print("\nExcel File Loaded Successfully")
print("Shape:", df_excel.shape)
print(df_excel.head())
```

```
Excel File Loaded Successfully
Shape: (715, 18)
   Sample   Age  Photos  # of Tags  Albums  Gender   Emp  Profile  Cover  \
0       1   57       7         27       5       1    1        1      0
1       2   42     531        241      19       0    0        0      0
2       3   42    1396        423      40       0    1        1      0
3       4   55     394        139      48       0    1        1      1
4       5   30     916        231      78       1    1        1      1

   Orientation  Relationship  Posts  Replies  Children  Likes  Edu  Events  \
0            1             1      1        0         3     21    0       0
1            1             1      5        1         3      9    0       1
2            1             1    186       31         2     43    1       1
3            1             1      6        4         2    187    0       4
4            1             1    412      752         1     34    0       2

   Friends
0       68
1      234
2      727
3      437
4      372
```

Compare All Three Datasets

```
print("\nTXT shape:", df_txt.shape)
print("CSV shape:", df_csv.shape)
print("Excel shape:", df_excel.shape)

print("\nTXT vs CSV same?:", df_txt.shape == df_csv.shape)
print("CSV vs Excel same?:", df_csv.shape == df_excel.shape)
```

```
TXT shape: (715, 31)
CSV shape: (715, 3)
Excel shape: (715, 18)

TXT vs CSV same?: False
CSV vs Excel same?: False
```

Use Excel Data for the Rest of the Analysis

```
df = df_excel.copy()
```

Data Examination

```
print("Column Names:")
print(df.columns)

print("\nTop 5 Rows:")
print(df.head())

print("\nBottom 5 Rows:")
print(df.tail())

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Column Names:
Index(['Sample ', 'Age', 'Photos', '# of Tags', 'Albums', 'Gender ', 'Emp',
       'Profile', 'Cover', 'Orientation', 'Relationship', 'Posts', 'Replies',
```

```
       'Children', 'Likes', 'Edu', 'Events', 'Friends'],
      dtype='object')

Top 5 Rows:
   Sample   Age  Photos  # of Tags  Albums  Gender   Emp  Profile  Cover  \
0       1   57       7         27       5       1    1        1      0
1       2   42     531        241      19       0    0        0      0
2       3   42    1396        423      40       0    1        1      0
3       4   55     394        139      48       0    1        1      1
4       5   30     916        231      78       1    1        1      1

   Orientation  Relationship  Posts  Replies  Children  Likes  Edu  Events  \
0            1             1      1        0         3     21    0       0
1            1             1      5        1         3      9    0       1
2            1             1    186       31         2     43    1       1
3            1             1      6        4         2    187    0       4
4            1             1    412      752         1     34    0       2

   Friends
0       68
1      234
2      727
3      437
4      372

Bottom 5 Rows:
     Sample   Age  Photos  # of Tags  Albums  Gender   Emp  Profile  Cover  \
710      711   25     499          4       0       0    1        1      1
711      712   22     232        101       4       1    0        1      0
712      713   22     327        329      29       1    1        1      0
713      714   24     219          0       8       1    1        1      1
714      715   19     465          0      22       0    1        1      0

     Orientation  Relationship  Posts  Replies  Children  Likes  Edu  Events  \
710            1             1     16       19         0     49    0      11
711            1             0    698      552         0    223    0       4
712            1             1    543      399         0    520    0       6
713            1             0      2        7         1     66    1       1
714            1             1      3        6         0    990    1       3

     Friends
710     1689
711      327
712      706
713      500
714      585

Data Types:
Sample          int64
Age             int64
Photos          int64
# of Tags       int64
Albums          int64
Gender          int64
```

Descriptive Statistics

```
print("\nDescriptive Statistics:")
print(df.describe())
```

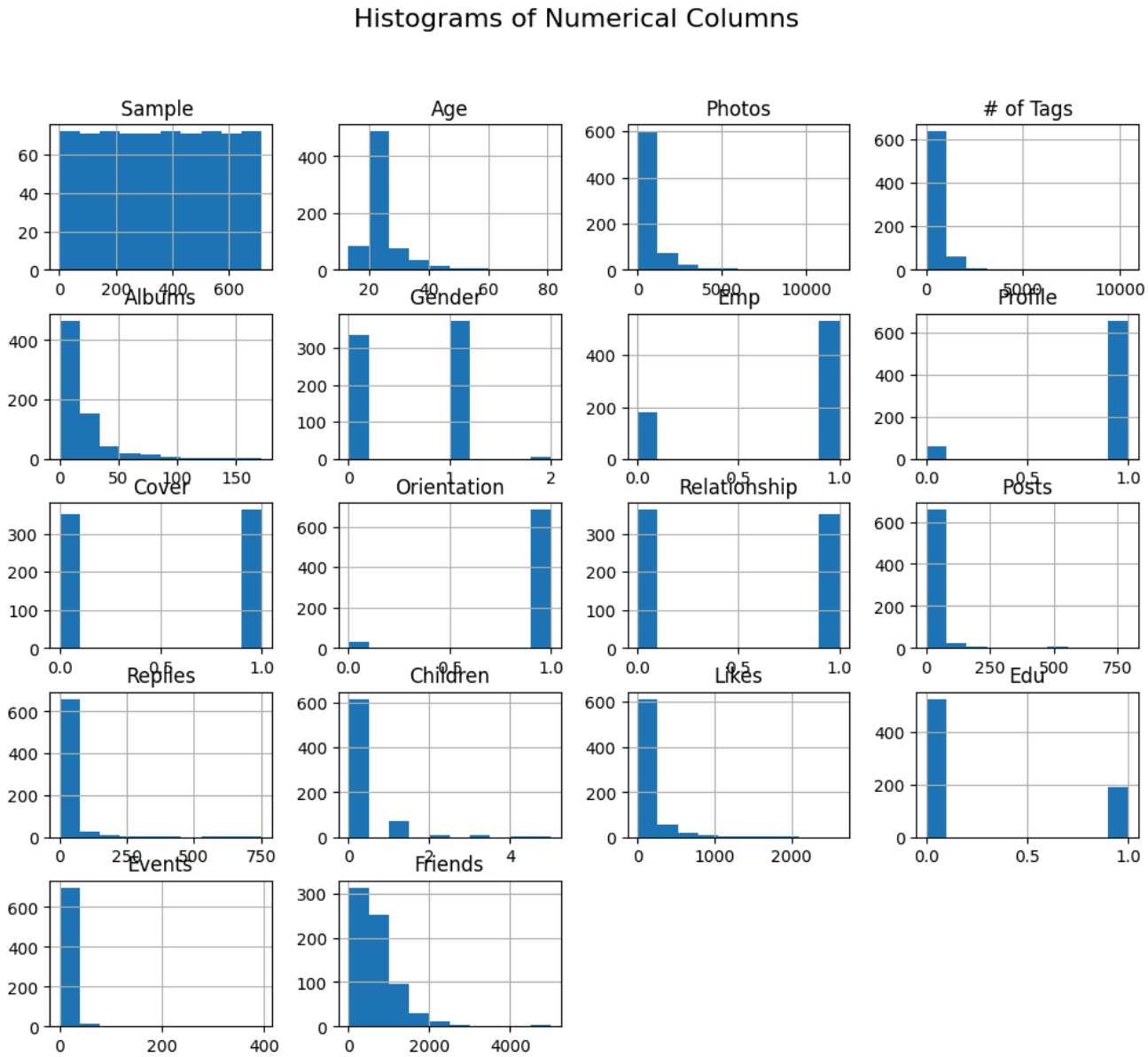
```
Descriptive Statistics:
          Sample          Age        Photos     # of Tags      Albums  \
count  715.000000  715.000000    715.000000    715.000000  715.000000
mean   358.000000   24.406993    723.914685    458.096503   18.952448
std    206.547008    6.969578   1265.112304    720.835481   22.502860
min      1.000000   13.000000      0.000000      0.000000    0.000000
25%    179.500000   21.000000    124.500000     75.500000    6.000000
50%    358.000000   22.000000    320.000000    250.000000   12.000000
75%    536.500000   25.000000    757.000000    566.500000   22.500000
max    715.000000   81.000000  11995.000000  10460.000000  172.000000

          Gender          Emp     Profile       Cover  Orientation  \
count  715.000000  715.000000  715.000000  715.000000   715.000000
mean     0.541259    0.745455    0.917483    0.507692     0.956643
std      0.517932    0.435910    0.275344    0.500291     0.203801
min      0.000000    0.000000    0.000000    0.000000     0.000000
25%      0.000000    0.000000    1.000000    0.000000     1.000000
50%      1.000000    1.000000    1.000000    1.000000     1.000000
75%      1.000000    1.000000    1.000000    1.000000     1.000000
```

```
max      2.000000    1.000000    1.000000    1.000000     1.000000

       Relationship       Posts     Replies    Children        Likes  \
count    715.000000  715.000000  715.000000  715.000000   715.000000
mean       0.492308   37.579021   34.816783    0.211189   144.812587
std        0.500291   90.437795   83.629158    0.651555   261.414781
min        0.000000    0.000000    0.000000    0.000000     0.000000
25%        0.000000    6.000000    5.000000    0.000000    21.000000
50%        0.000000   15.000000   14.000000    0.000000    59.000000
75%        1.000000   29.500000   29.000000    0.000000   145.000000
max        1.000000  794.000000  752.000000    5.000000  2619.000000

               Edu      Events      Friends
count  715.000000  715.000000   715.000000
mean     0.267133    8.853147   695.853147
std      0.442772   24.002672   599.527946
min      0.000000    0.000000     3.000000
25%      0.000000    1.000000   302.000000
50%      0.000000    4.000000   563.000000
75%      1.000000    9.000000   893.500000
max      1.000000  396.000000  4999.000000
```

Data Visualization

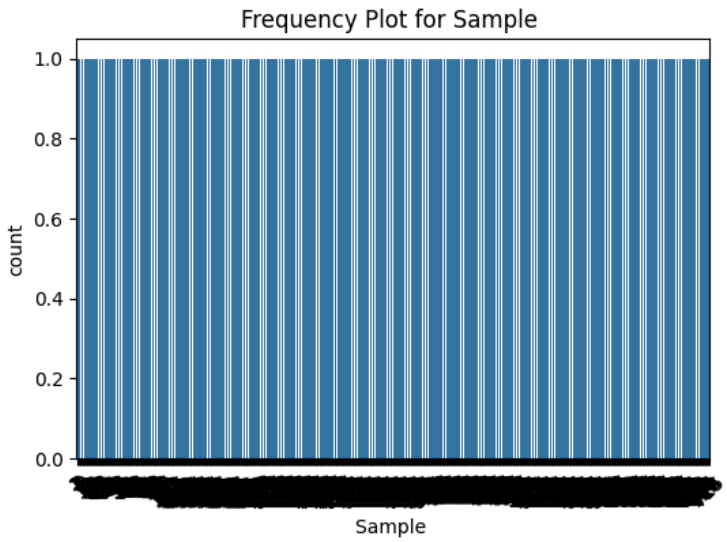
Histograms for All Numerical Columns

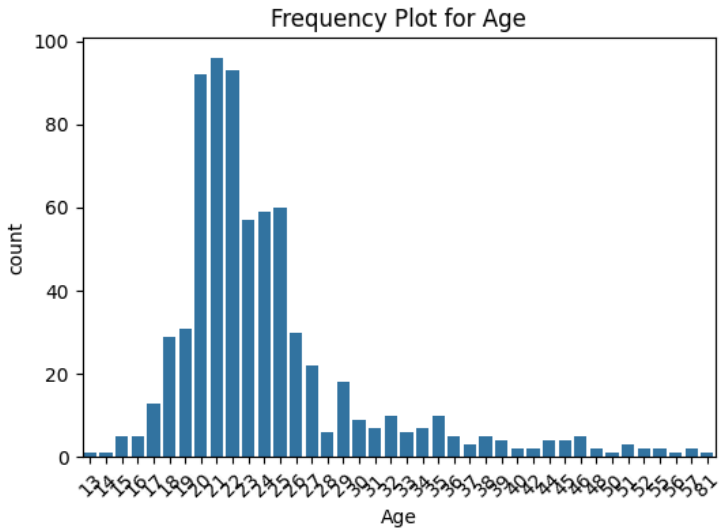
```
df.hist(figsize=(12, 10))
plt.suptitle("Histograms of Numerical Columns", fontsize=16)
plt.show()
```

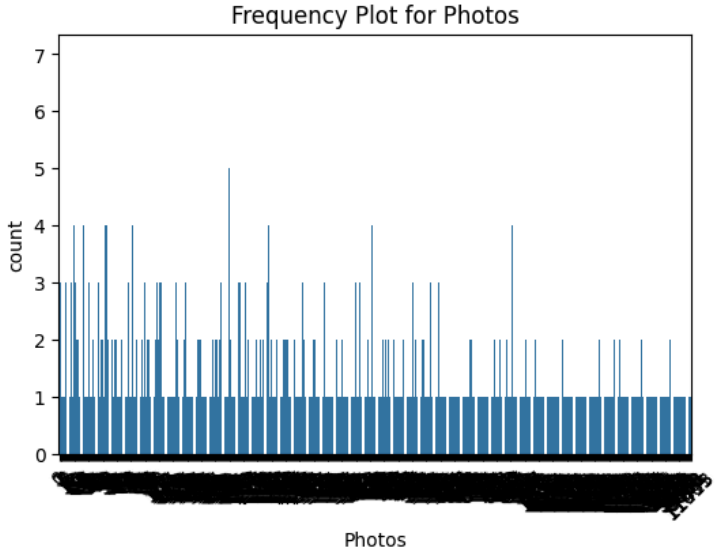


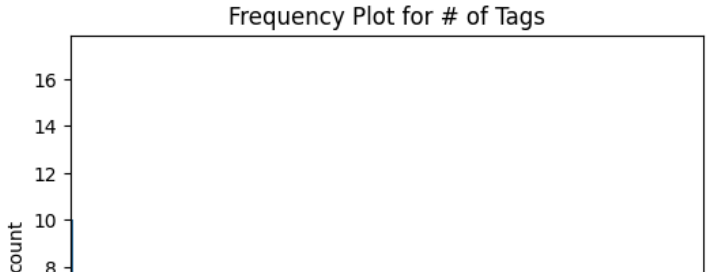
Frequency Plots for All Integer/Binary Columns

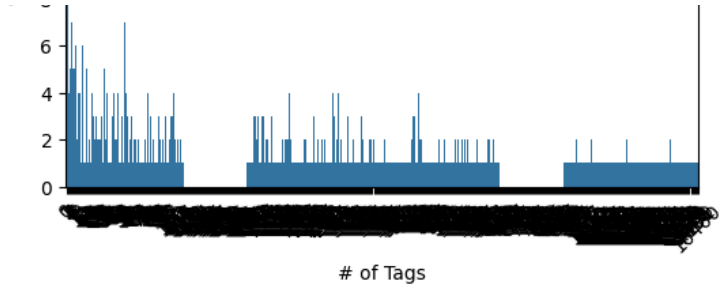
```
categorical_cols = df.select_dtypes(include="int64").columns

for col in categorical_cols:
    plt.figure(figsize=(6, 4))
    sns.countplot(x=df[col])
    plt.title(f"Frequency Plot for {col}")
    plt.xticks(rotation=45)
    plt.show()
```


Frequency Plot for Sample
count
1.0
0.8
0.6
0.4
0.2
0.0
Sample

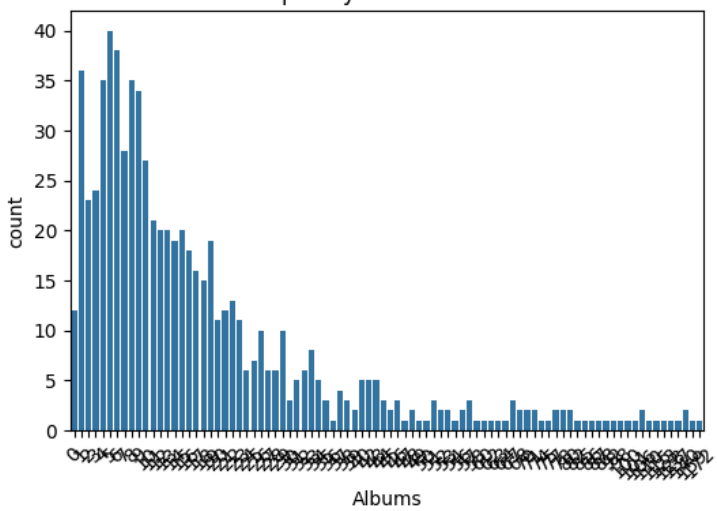

Frequency Plot for Age
count
100
80
60
40
20
0
Age


Frequency Plot for Photos
count
7
6
5
4
3
2
1
0
Photos

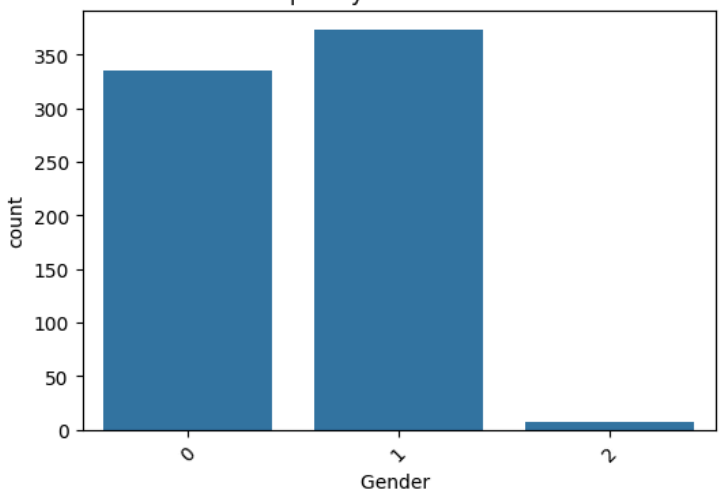

Frequency Plot for # of Tags
count
16
14
12
10
8



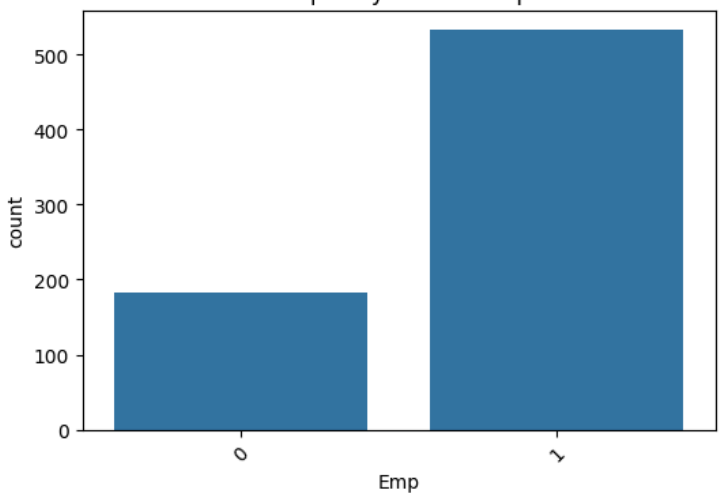
Frequency Plot for Albums



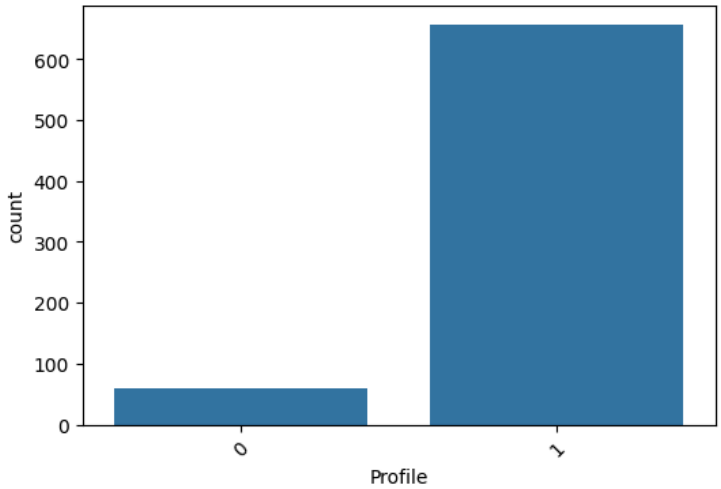
Frequency Plot for Gender

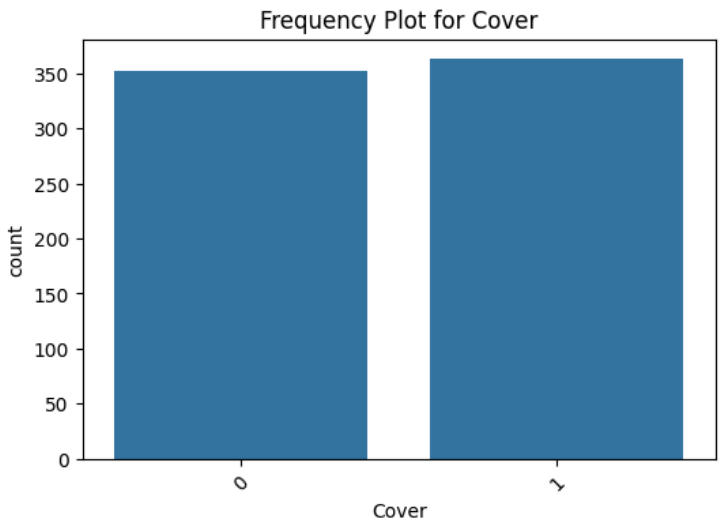


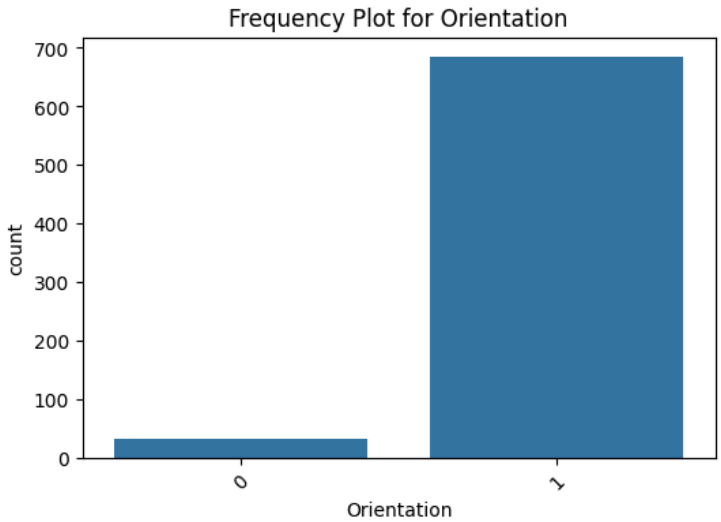
Frequency Plot for Emp

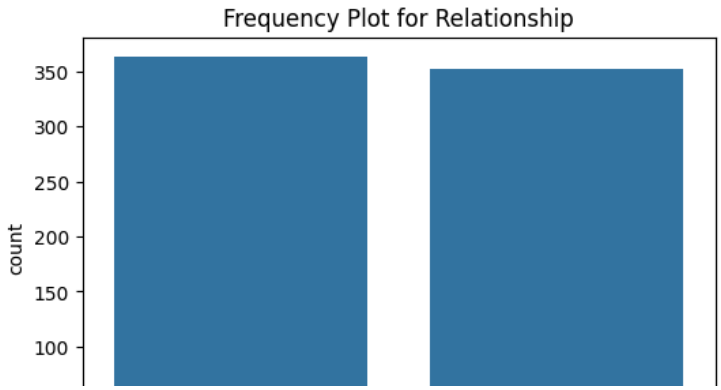


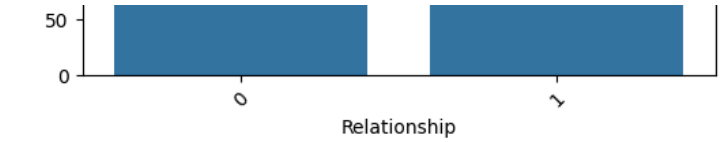
Frequency Plot for Profile

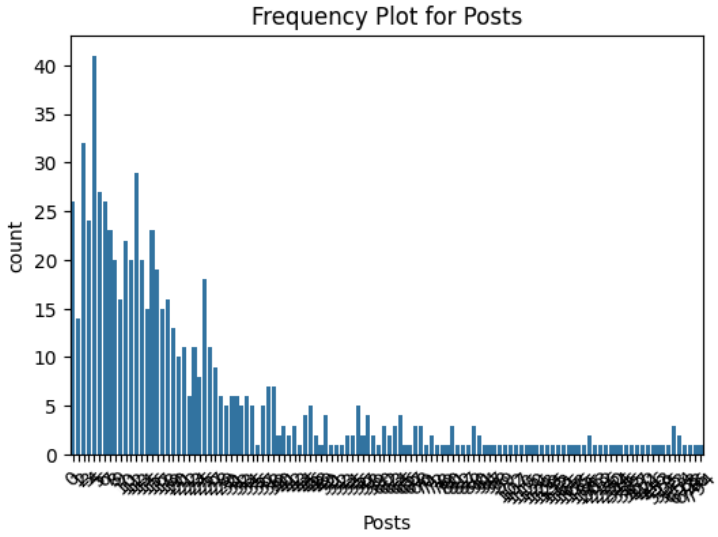

count
Profile
0
1

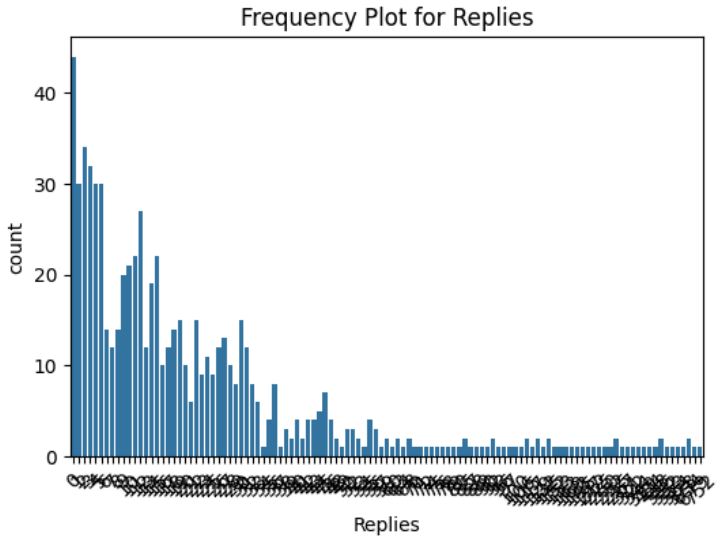

Frequency Plot for Cover
count
Cover
0
1

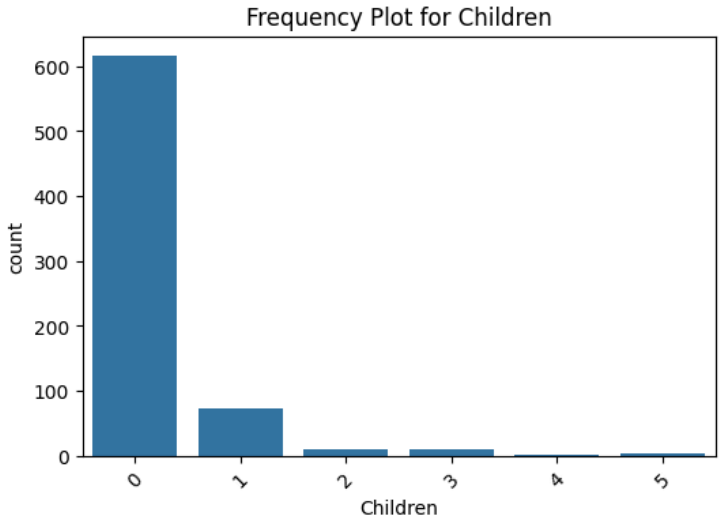

Frequency Plot for Orientation
count
Orientation
0
1

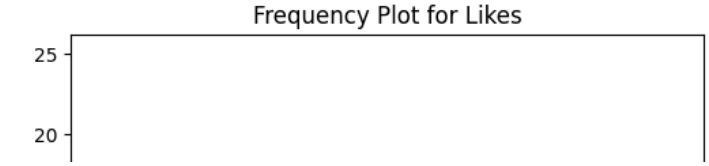

Frequency Plot for Relationship
count


50
0
0
1
Relationship


Frequency Plot for Posts
count
Posts


Frequency Plot for Replies
count
Replies


Frequency Plot for Children
count
0
1
2
3
4
5
Children


Frequency Plot for Likes